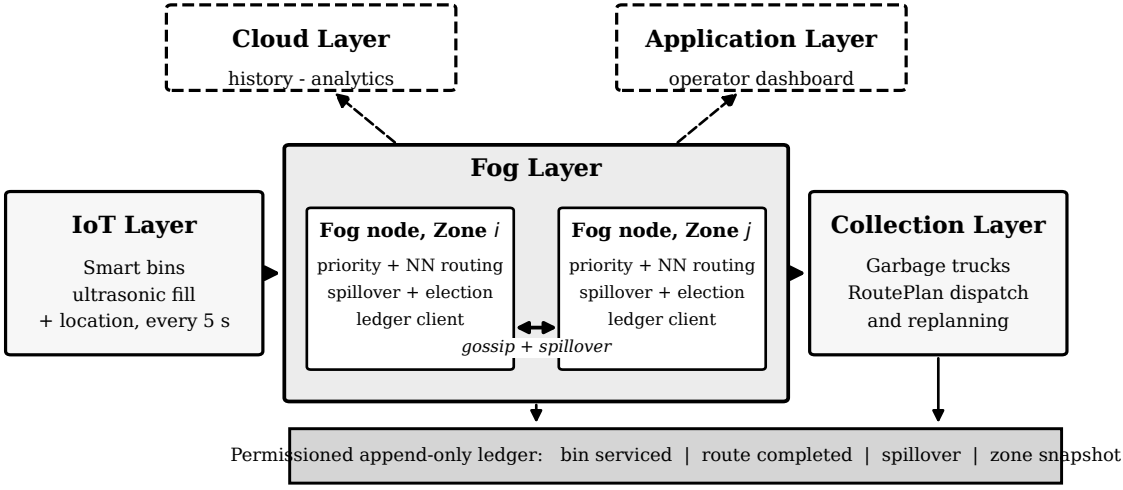
Cloud Layer
history - analytics
Application Layer
operator dashboard
Fog Layer
IoT Layer
Smart bins
ultrasonic fill
+ location, every 5 s
Fog node, Zone i
priority + NN routing
spillover + election
ledger client
Fog node, Zone j
priority + NN routing
spillover + election
ledger client
gossip + spillover
Collection Layer
Garbage trucks
RoutePlan dispatch
and replanning
Permissioned append-only ledger: bin serviced | route completed | spillover | zone snapshot
real-time critical path (no cloud round trip)
asynchronous flow, off the decision path